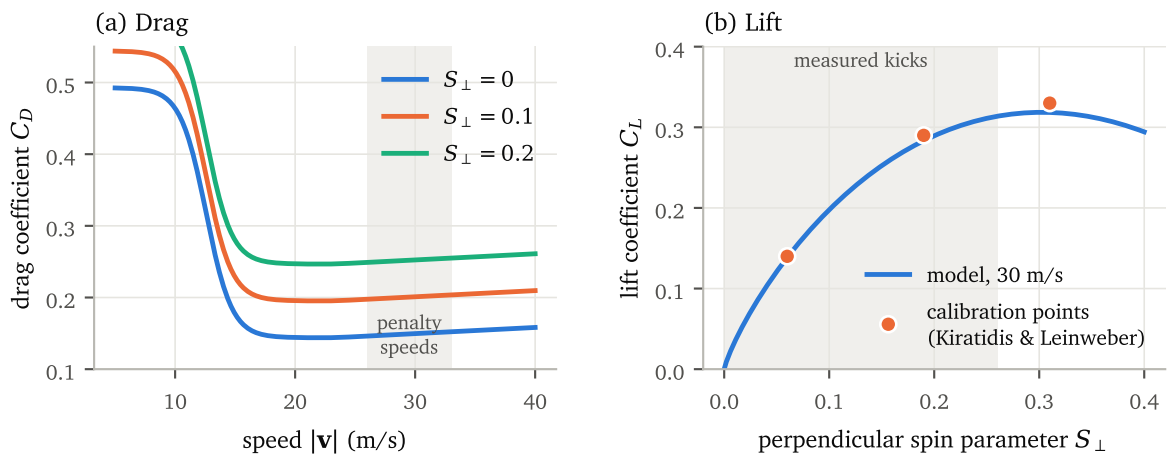
(a) Drag
drag coefficient C_D
0.5
0.4
0.3
0.2
0.1
10
20
30
40
speed $|\mathbf{v}|$ (m/s)
$S_\perp = 0$
$S_\perp = 0.1$
$S_\perp = 0.2$
penalty
speeds
(b) Lift
lift coefficient C_L
0.4
0.3
0.2
0.1
0.0
0.0
0.1
0.2
0.3
0.4
perpendicular spin parameter $S_\perp$
measured kicks
model, 30 m/s
calibration points
(Kiratidis & Leinweber)